

"ALIEN" SPECIAL-SHAPE HOT AIR BALLOON

BUILD PACK

2,610 m3 / 92,171 cu ft - 20 gores x 17 panel rows - green ripstop, black features

Volume	2,610 m3 / 92,171 cu ft
Envelope height	18,338 mm / 60 ft 2 in (~20,620 mm over basket)
Envelope width	17,887 mm / 58 ft 8 in
Mouth / parachute hole	O3,988 mm / O5,000 mm (seal overlap 360 mm)
Gore length	22.83 m / 74 ft 11 in - 17 rows, NOMEX + A..P
Basis	master pattern set - full-size master patterns (no scaling)
Seam allowance	25 mm (1 in), 5/8 in double French fell
Hole + appendage tape	Class 3 1/2 in (12.7 mm) load tape - every panel hole, the eye rim, the antenna base

FEATURES

Eyes	two black ram-air pods - gores 18-20 / 2-4, rows F-J, ~1.3 m proud - skin sewn STRAIGHT to the envelope over rim tape
Feed holes	ENVELOPE only, O450 cut / O500 finished, hemmed over Class 3 1/2 in tape, three per pod: 20/2: panel H +450 mm; 19/3: panel H +910 mm; 18/4: panel i +850 mm
Mouth	black crescent applique, row C across gores 20/1/2, three per-panel pieces applied FLAT before gore assembly
Antennae	two x 3.75 m (2.25 m cone + O1.5 m red ball), row N gores 18 & 4, 35 deg lean - base is a DOUBLED panel N
Turning vents	overlap type at seams 15/16 and 16/17 flanking gore 16 (left side, face up), rows i-J
Black areas	NOMEX bottom row, scoop, eye pods, mouth crescent, antenna cones

SHOP LIMITS THIS PACK IS BUILT TO

PRINTING - read before you print anything

Cloth 1,524 mm / 60 in bolt - no cut piece is wider than 1,474 mm.

Table 3,658 mm / 12 ft - no cut piece is longer than 3,600 mm.

Cut envelope panels with the ROW HEIGHT across the bolt and the

gore width along the roll. Row heights are 1,357-1,411 mm - the

master patterns were already sized for a 60 in bolt.

The eye pod skin is the only piece too big for any bolt: it is cut

as strips, and every strip piece then prints 1:1.

All 1:1 sheets print at 100 % - NEVER 'fit to page' or 'shrink

oversized pages'.

Measure the CALIBRATION LINES on every 1:1 sheet before you cut.

Every 1:1 sheet has one side under 1,500 mm so it fits a 60 in roll.

The eye skin cannot be cut whole, but each PIECE prints 1:1 - see

alien-balloon-eye-patterns.pdf.

Solid = CUT. Dotted = SEW (finished). Red = feature marks.

BUILD PACK - FILE INDEX

1:1 = PATTERNS ONLY, print actual size on a 60 in roll - every page carries a calibration line. REF = reference reading, any printer. CAD = DXF.

START HERE

alien-balloon-00-BUILD-PACK-INDEX.pdf	REF	this document - specs, index, build order
alien-balloon-spec-summary.pdf	REF	all key dimensions, metric + English
alien-balloon-final-concept.html	REF	concept drawing: front, rear, antenna detail
alien-balloon-3d.html	REF	interactive 3D model (open in a browser)

LAYOUT & PLANNING

alien-balloon-layout-diagram.html	REF	gore x row grid with every feature located
alien-balloon-panel-map.html / .csv	REF	which panel carries which feature
alien-balloon-gore-layout.pdf	REF	21 pages: all gores + one page per gore
alien-balloon-panel-dimensions.html / .csv	REF	finished and cut sizes, every panel
alien-balloon-panel-loft.html / .csv	REF	panel width at 0/25/50/75/100 % height
alien-balloon-sewing-schedule.html	REF	seam specs, tape schedule, assembly sequence
alien-balloon-build-log.pdf	REF	6 sign-off sheets - print and use on the floor

PANEL TEMPLATES (cut the envelope)

alien-balloon-panels-A-B-Nomex.pdf	1:1	panels A, B, NOMEX nested
alien-balloon-panels-C-D.pdf	1:1	panels C, D - C carries the mouth outline
alien-balloon-panels-E-F.pdf	1:1	panels E, F
alien-balloon-panel-G/H/I/J/K/L.pdf	1:1	six single-panel sheets - eyes, feed holes, vent note
alien-balloon-panels-M-N.pdf	1:1	panels M, N - cut 22 of N (2 are the antenna doublers)
alien-balloon-panels-O-P.pdf	1:1	panels O, P
alien-balloon-panels-grouped.pdf	1:1	all five grouped sheets in one file

APPENDAGES

alien-balloon-eye-build.pdf	REF	ILLUSTRATED build procedure - 9 sheets, start here for the eyes
alien-balloon-eye-patterns.pdf	1:1	14 skin cutting templates, one piece per page
alien-balloon-eyes.pdf	REF	spec, marking + skin ordinates, piecing plan, cut list
alien-balloon-mouth.pdf	1:1	3 pages, one crescent piece per gore
alien-balloon-mouth-instructions.pdf	REF	flat-applique method, exact edges
alien-balloon-antennae.pdf	1:1	PATTERNS ONLY - cone panel, ball gore, base-opening template
alien-balloon-antenna-instructions.pdf	REF	detail specification, ball sewing, cone sewing + fitting
alien-balloon-turning-vents.pdf	REF	vent placement and assembly
alien-balloon-vent-panels.pdf	REF	which panels to cut for the vents

TOP & BOTTOM / CAD

alien-balloon-parachute.pdf	REF	parachute data and full rigging lengths
alien-balloon-scoop.pdf	REF	scoop, mouth rigging, flying-wire tables
alien-balloon-panels.dxf	CAD	full 2D envelope development (+ png preview)
alien-balloon-panel-templates.dxf	CAD	17 panels on a grid, cut + sew layers
alien-balloon-appendage-patterns.dxf	CAD	eyes, mouth, antenna pieces

BUILD ORDER

follow in sequence - each step names the sheet to work from

1	PRINT & VERIFY	Print all 1:1 sheets at 100 %. Measure the calibration lines on each - both directions.
2	CUT PANELS	Cut 20 of each panel, plus 2 EXTRA panel N for the antenna doublers. Lay the ROW HEIGHT across the 60 in bolt and the gore width along the roll. Mark each with its letter and an UP arrow - panels are not symmetric top to bottom.
3	EXTRA PANELS	Cut 2 extra panel J and 2 extra panel i with the vent markings for the turning vents. The 2 extra panel N (antenna doublers) go on the inside face of gores 18 & 4 and are caught into all four seams during gore assembly.
4	MOUTH APPLIQUE	Applique the three crescent pieces FLAT onto panel C of gores 20, 1 and 2. Crescent edges exact; seam-end extensions left free.
4b	EYE POD SKIN	Work from alien-balloon-eye-build.pdf. Cut the 14 skin pieces per pod 1:1 from alien-balloon-eye-patterns.pdf, sew sections then strips, then check the assembled outline against the whole-shape ordinates. The envelope panels are the pod's base - there is no backing layer.
5	FEED HOLES	Cut the eye feed holes (20/2: panel H +450 mm; 19/3: panel H +910 mm; 18/4: panel i +850 mm) and the antenna holes (row N, gores 18 & 4, O575/O625).
6	BUILD GORES	Join panels into 20 vertical gores, felling row seams bottom-up (NOMEX first).
7	JOIN GORES	Join gores 1-20 over the vertical load tapes; catch the mouth-crescent extensions in the 20/1 and 1/2 fells.
8	TURNING VENTS	Fit both vent units at seams 15/16 and 16/17, rows i-J: lips hemmed over tape, 70 mm overlap (J outside i), triangle sew lines.
9	EYE PODS	Mark the footprint outline on the envelope, sew 22.8 m of Class 3 1/2 in tape to it (caught into the gore load tapes), then ease the skin on - 226 mm oversize, 6 % rim ease over 16 match points. Load-test 25 kg at four points.
10	ANTENNAE	Cones and balls onto the doubled panel N of gores 18 & 4 - the doubler is a second panel caught into all four seams, so it must go in at gore assembly, not now. 35 deg outboard lean.
11	MOUTH & APERTURE	Hem the mouth (O3,988 mm) over webbing and the aperture (O5,000 mm) over tape. Fit the black scoop.
12	PARACHUTE	Set the parachute with the top-opening reference line (360 mm seal overlap). Rig confluence 3,600 mm, centering 3,639 mm (trim toward 3,598).
13	LINES & WIRES	Crown line 31.0 m, red line 34.0 m (+3 m each). Flying wires per the scoop sheet table for your burner frame.
14	INSPECT & SIGN OFF	Seam pull tests, vent function, pod inflation, parachute seating. Complete all six build-log sheets.

Airworthiness: this pack documents the design and construction only.

Registration, inspection and any airworthiness approval for a home-built envelope are the builder's responsibility - involve your national authority and an experienced balloon repair station before the first inflation.